SYNTHETIC scalar PTT

a
Injected law
PTT (s)
BP (mmHg)
b
Training
loss
epoch
train
val
c
Audit, α = 1
pred. BP (mmHg)
PTT shift (ms)
d
Accuracy is blind to α
score
α (PTT weight)
accuracy
linear probe
audit (scaled)

SYNTHETIC waveforms

e
Synthetic ECG and PPG
ECG
PPG
time (s)
f
Injected law, α = 1
BP (mmHg)
injected PTT (ms)
-0.22 mmHg/ms
g
Transformer training
RMSE (mmHg)
mean
oracle
train
val
epoch
h
Detector vs. ground truth
measured − 400 ms offset
injected PTT (ms)
r = 0.998
100% measurable
i
Audit recovers the law
audit slope (mmHg/ms)
r = -1.000
injected
α (PTT weight)

REAL VitalDB

j
VitalDB ECG and PPG
ECG
PPG
time (s)
k
Transformer training
DBP MAE (mmHg)
mean (9.5)
train
val
epoch
l
BP distribution
144 test subjects
mmHg
SBP train
SBP test
DBP train
DBP test
m
Linear probes
probe R^2
PAT
cardiac period
embed
L1
L2
L3
layer
n
Sign test (142 subjects)
expected
textbook
100% neg.
transformer
33% neg.
inverted
0% neg.
per-subject slope (mmHg/ms)